Daily Algebra Drill #7

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Weighted AM–GM · Vieta jumping · Power mean inequality · Infinite products via telescoping.

Section A Rapid Recognition

1. Factorise: $x^5 - x$.

Ans: $x(x-1)(x+1)(x^2+1)$

2. Given $ab + bc + ca = 4$ and $a + b + c = 3$, find $(a + b + c)^2 - (a^2 + b^2 + c^2)$.

Ans: 8

3. Without a calculator: $7^2 - 5^2 + 3^2 - 1^2$.

Ans: 32

4. Roots of $x^2 - 8x + 3 = 0$ are p, q. Find $\left(p - \frac{1}{p}\right)\left(q - \frac{1}{q}\right)$.

Ans: -16

5. Simplify: $\dfrac{(2n)!}{(2n-1)!} - \dfrac{n!}{(n-1)!}$.

Ans: n

6. Factorise: $27x^3 - 8$.

Ans: $(3x-2)(9x^2+6x+4)$

7. Given $t = x - \frac{1}{x}$, write $t^2 + 4$ in terms of x.

Ans: $\left(x + \frac{1}{x}\right)^2$

8. Evaluate: $\log_2 8 + \log_2 4 - \log_2 32$.

9. Given $f(x) = 2x + 1$ and $g(x) = x^2 - 1$, find $f(g(x)) - g(f(x))$.

10. Simplify: $\dfrac{a^2 - b^2}{a - b} - \dfrac{a^2 - c^2}{a - c}$.

Section B Manipulation Drills

1. Evaluate the telescoping product $\prod_{k=2}^{n}\left(1 - \frac{1}{k}\right)$ for general n and at $n = 100$.

2. Simplify: $\dfrac{x^2+3x-10}{x^2-x-2} \cdot \dfrac{x^2+x-2}{x^2+6x+5}$.

Ans: $\dfrac{(x+1)^2}{(x+2)(x-1)}$

3. Find $a^3+b^3+c^3$ given $a+b+c=2$, $a^2+b^2+c^2=6$, $abc=-2$.

Ans: 8

4. Write x^4+4 as a product of two quadratics with integer coefficients.

Ans: $(x^2+2x+2)(x^2-2x+2)$

5. Find $f^{-1}(x)$ for $f(x) = \dfrac{2x+1}{x-1}$ and verify $f(f^{-1}(x)) = x$.

Ans: $f^{-1}(x) = \dfrac{x+1}{x-2}$

6. Prove: the product of four consecutive integers is always one less than a perfect square.

Ans: $(n^2+n-1)^2 - 1$

7. Simplify: $\dfrac{1}{1+x} + \dfrac{1}{1-x} + \dfrac{2}{1+x^2} + \dfrac{4}{1+x^4}$.

Ans: $\dfrac{8}{1-x^8}$

8. Roots α, β, γ of $x^3 + px + q = 0$. Express $\alpha^3 + \beta^3 + \gamma^3$ in terms of p and q without Newton's identity.

Ans: $-3q$

9. Prove: if $a + b + c = 0$ then $(a^2 + b^2 + c^2)^2 = 2(a^4 + b^4 + c^4)$.

Ans: Proved below.

10. Solve: $\dfrac{x+1}{x-2} + \dfrac{x-2}{x+1} = \dfrac{10}{3}$.

Ans: $x = \frac{7}{2}$ and $x = -\frac{5}{2}$

Section C Substitution & Structure

1. Prove $\dfrac{x}{3} + \dfrac{2y}{3} \geq x^{1/3}y^{2/3}$ for $x, y > 0$ using weighted AM–GM.

Ans: Proved below.

2. Find all real pairs (x, y) with $x + y = 1$ and $x^4 + y^4 = \frac{41}{8}$.

Ans: $\left(\frac{3}{2}, -\frac{1}{2}\right)$ and $\left(-\frac{1}{2}, \frac{3}{2}\right)$

3. Prove $\sqrt{\dfrac{a^2+b^2}{2}} \geq \dfrac{a+b}{2} \geq \sqrt{ab} \geq \dfrac{2ab}{a+b}$ for $a, b > 0$.

Ans: All three inequalities proved below.

4. Solve: $27^x + 27^{1-x} = 28$.

Ans: $x = 1$ and $x = 0$

5. For $a, b, c > 0$ with $a + b + c = 1$, prove $\dfrac{1}{a} + \dfrac{1}{b} + \dfrac{1}{c} \geq 9$.

Ans: Proved below.

6. Evaluate $3^3 + 4^3 + \cdots + 10^3$ using $\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$.

Ans: 3016

7. Find all real solutions to $|x^2 - 4| = |x - 2| \cdot |x + 2|$.

Ans: All real x.

8. Given $x, y > 0$ with $x + y = 4$, find the maximum of $x^2 y$.

Ans: Maximum is $\frac{256}{27}$ at $x = \frac{8}{3}$, $y = \frac{4}{3}$.

Section D Challenge (SMC / BMO1 difficulty)

1. **Vieta jumping:** Suppose a and b are positive integers with $a^2 + b^2 = k(ab+1)$. Show $k = 4$ is achieved by $(a, b) = (8, 2)$, and explain the descent step.

Ans: (a) $(a, b) = (8, 2)$ gives $k = 4$. (b) Vieta jumping establishes no new solutions beyond the obvious family.

2. Find all polynomials $p(x)$ satisfying $(x-1)p(x+1) = (x+2)p(x)$.

Ans: $p(x) = cx(x-1)(x+1)$ for any constant c.

3. For $a+b+c=1$, $a,b,c>0$, find $\min\left(\dfrac{a}{1-a}+\dfrac{b}{1-b}+\dfrac{c}{1-c}\right)$.

Ans: $\frac{3}{2}$

4. Evaluate $\displaystyle\prod_{n=2}^{\infty}\frac{n^3-1}{n^3+1}$.

Ans: $\frac{2}{3}$

5. Find all integer solutions to $x^3 + y^3 = (x + y)^2$.

Ans: $(n, -n)$ for all $n \in \mathbb{Z}$, and $(1, 0), (0, 1), (1, 2), (2, 1), (2, 2)$